

Musthofa Joko Anggoro

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RESEARCH INTERESTS

My thesis was my first serious encounter with hardware design — building a working CPU in VHDL and seeing it run on real silicon. That got me curious about a harder problem: how do you map AI workloads onto reconfigurable hardware efficiently? I want to work on that question, and I do not have all the tools to answer it yet. That is why I am applying for a PhD.

EDUCATION

University of Indonesia (UI) — Faculty of Computer Science

Jakarta, Indonesia

Bachelor of Computer Science, Information Systems

Aug 2022 – Aug 2026 (Expected)

- **GPA: 3.84 / 4.00**

- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

Nanyang Technological University (NTU) & ITS

Singapore & Surabaya

INSPIRASI-NTU Summer Programme 2025 — LPDP Scholarship Awardee

July 2025

- Applied software engineering and systems principles to sustainable development; produced a technical report proposing a tiered smart-port automation framework with IoT sensor networks and renewable energy integration

UNDERGRADUATE THESIS

VHDL-93 Implementation of the Mano Register-Transfer Microoperation Model on FPGA

Supervisor: Prof. Petrus Mursanto, University of Indonesia *Status:* Completed — *Paper in preparation for ICACSYS 2026*

- Designed and implemented the Mano register-transfer computer model as a complete, hierarchically structured VHDL-93 (IEEE 1076-1993) datapath on a **Xilinx Spartan-3AN** FPGA
- Conducted a systematic **gap analysis** mapping 5 undefined textbook constraints to concrete RTL design decisions, bridging theory and physical hardware implementation
- Implemented a **three-tier component hierarchy**: leaf-level primitives (`reg_n`, ALU, shifter) → functional units → full system integration (`fpga_top`)
- Verified all modules via **ISim functional simulation** testbenches and board-level testing on physical hardware
- Achieved **timing closure at 50 MHz** on the Spartan-3AN device (critical path: 17.05 ns) using XST synthesis engine
- Demonstrated live operation via PS/2 keyboard input and **HD44780 LCD display** output; programmed via JTAG
- *Toolchain:* Xilinx ISE 14.x · XST · ISim · JTAG · VHDL-93

PUBLICATIONS

- Musthofa Joko Anggoro, Petrus Mursanto. “VHDL-93 Implementation of the Mano Register-Transfer Microoperation Model on an FPGA.” *Proceedings of the International Conference on Advanced Computer Science and Information Systems (ICACSYS) 2026*. [In preparation]

TECHNICAL SKILLS

HDL & Digital Design	VHDL-93, RTL synthesis, place-and-route, functional simulation, testbench writing
FPGA Tools	Xilinx ISE 14.x, XST, ISim, JTAG; Device: Spartan-3AN (XC3S700AN)
Programming	Python, Java, C, Go, TypeScript
Machine Learning	Scikit-learn, supervised/unsupervised learning, model evaluation
Infrastructure	Docker, Kubernetes, CI/CD, Linux/Unix, Git